

Dimension	Cloud-centric architecture	Edge-native architecture
Response latency	100-500ms round-trip time; unsuitable for real-time applications requiring sub-10ms response	1-10ms response times; enables real-time decision-making at the source
Bandwidth economics	Continuous streaming of raw data can cost \$40+ per device monthly for high-resolution sensors; costs scale linearly with device count	90% bandwidth reduction through local processing and selective data transmission; costs remain relatively flat as fleet grows
Data privacy and compliance	Sensitive data must traverse public networks and reside in third-party infrastructure; complex compliance requirements	Data processed and retained locally; sensitive information never leaves premises; simplified compliance posture
Operational resilience	Complete dependency on network connectivity and cloud availability; single point of failure	Autonomous operation during network outages; graceful degradation; continues functioning independently
Scalability economics	Infrastructure costs grow proportionally with data volume and processing needs; potential for runaway costs	Processing is distributed across edge devices; cloud costs grow sub-linearly; predictable cost structure
Decision making	Centralised analysis enables global optimisation but requires complete data visibility	Local autonomy with federated intelligence; can implement hierarchical decision-making
Initial complexity	Lower barrier to entry; well-understood patterns and abundant expertise	Higher initial architectural complexity requires distributed systems expertise
Maintenance	Centralised updates and monitoring; simpler operational model	Distributed device management requires robust OTA update mechanisms

Table 1: Cloud vs edge: A comparative analysis

separates successful deployments from failed experiments.

The architecture blueprint

Designing a production-grade edge AI system requires careful orchestration of multiple technology layers, each serving distinct functions while maintaining seamless integration. The architecture must balance competing concerns: computational efficiency on resource-constrained devices, robust communication in unreliable network conditions, security against evolving threats, and operational manageability across potentially thousands of distributed nodes. The following architectural framework represents battle-tested patterns that have proven successful in enterprise deployments across various industries, including manufacturing, retail, healthcare, and

smart city applications.

Layer 1: Device and sensor layer — the physical foundation: The device and sensor layer is the backbone of an edge AI system, integrating digital intelligence with the physical world. It includes sensors to capture environmental data, edge computing hardware for processing, and the operating system for runtime. Hardware choices have a significant impact on power consumption, processing capabilities, security, and operational costs. Common platforms include the Raspberry Pi 4 for general computing, the NVIDIA Jetson Nano for GPU-accelerated applications, the ESP32 for low-power wireless use, and the Arduino for basic sensor tasks.

Layer 2: Communication and protocol layer - the nervous system: This layer functions as the nervous

system of edge AI, enabling data flow between devices, gateways, and the cloud. MQTT is the standard for IoT messaging, with Eclipse Mosquitto as a robust broker. Its publish-subscribe model simplifies connections and ensures reliable message delivery. For resource-constrained devices, CoAP provides a lightweight alternative.

Layer 3: Edge computing framework - the coordination layer: EdgeX Foundry addresses the complexity of IoT by providing a microservices framework that offers vendor-neutral abstractions for connectivity, data normalisation, and data export. Its architecture allows for component swapping without requiring redesign of the entire system. The device service simplifies diverse protocols, while the central data service ingests telemetry for other services.